Mitigating Shortcut Learning in Brain Tumour MRI Classification

Riya Basak
University of Hertfordshire
BSc Computer Science (Hons) with Artificial Intelligence

BSc final-year AI project

Abstract

Medical image classifiers can achieve high benchmark accuracy while relying on non-pathological shortcut cues. This work investigates whether pathology-guided hybrid CNN–Transformer models can reduce this risk in four-class brain tumour Magnetic Resonance Imaging (MRI) classification without using segmentation masks. Two project-specific guidance modules are proposed: Pathology-Focused Disentanglement (PFD), which learns a soft tumour-oriented spatial mask from CNN features, and Guided Semantic Token Evolution (GSTE), which uses this mask to guide transformer token weighting. Four models were evaluated: Hybrid A, Hybrid B, Ablation A, and Ablation B. Hybrid A applies weaker guidance to CNN-derived 7×7 tokens, while Hybrid B applies stronger guidance to raw-image patch tokens and the CNN descriptor. A leakage-aware benchmark was prepared from the Kaggle Brain Tumor MRI dataset using duplicate removal, fixed train/validation/test splits, and deterministic preprocessing. Evaluation combined held-out classification metrics with Grad-CAM++, Attention Rollout, and Monte Carlo dropout uncertainty. All models achieved high test accuracy, from 0.9852 to 0.9922. Ablation B achieved the strongest numerical performance, while Hybrid B produced the clearest tumour-centred Grad-CAM++ evidence in the reported cases. The results suggest that PFD-GSTE improved evidence focus more clearly than classification accuracy, supporting partial mitigation of shortcut learning rather than a complete solution.

Keywords: brain tumour MRI; shortcut learning; medical image classification; CNN–Transformer hybrid; pathology guidance; Grad-CAM++; Attention Rollout; uncertainty estimation.

1 Introduction

A central difficulty in medical image classification is that predictive success does not guarantee meaningful visual reasoning. A model may classify images correctly while relying on shortcut cues such as background structure, scanner-specific artefacts, preprocessing traces, or dataset-specific regularities rather than pathology itself (Geirhos et al., 2020; Roberts et al., 2021). In clinical imaging, this distinction matters because a model that performs well on one benchmark may fail when acquisition conditions, patient populations, or imaging protocols change.

Brain tumour MRI classification is a useful setting for studying this problem. Tumour subtypes can overlap visually, single two-dimensional slices may not contain the full anatomical context, and public aggregated datasets can contain hidden duplication or acquisition bias. This work therefore studies not only whether a model is accurate, but whether its evidence appears more tumour-centred.

The proposed method introduces two guidance modules into a hybrid CNN–Transformer classifier. Pathology-Focused Disentanglement (PFD) learns a soft spatial mask from CNN features. Guided Semantic Token Evolution (GSTE) uses this mask to guide transformer tokens. The broader hypothesis is simple: if the model is encouraged to route more attention through tumour-relevant regions, then its explanations may become less background-driven even when segmentation masks are unavailable.

2 Method Overview

2.1 Benchmark construction

The study uses the Kaggle Brain Tumor MRI dataset for four-class classification: glioma, meningioma, pituitary, and no tumour (Nickparvar, 2021). The dataset was prepared through an offline preprocessing pipeline. Exact duplicates were removed using SHA1 hashing, with held-out test-set protection. Kaggle Training was split into train and validation sets using stratified sampling, while Kaggle Testing was preserved as the final held-out test set.

The final prepared dataset contained 4353 training images, 1089 validation images, and 1284 test images. All images were corrected for orientation, tightly cropped to reduce black background, resized to 224×224, converted to RGB, and read through fixed CSV split files. Augmentation was applied only during training.

2.2 Architecture

Each model combines a ResNet50V2 CNN branch with an RViT-inspired transformer branch. The CNN branch provides local feature extraction, while the transformer branch models wider spatial relationships and rotated image views. The two branch descriptors are merged through late fusion before four-class prediction.

For an input image x, the CNN branch produces a feature map $F(x)$. PFD learns a soft mask:

$$M(x) = \sigma(g(F(x))),$$

where g is a learned projection and σ is the sigmoid function. The guided feature map is then:

$$F_{\text{guided}}(x) = F(x) \odot M(x).$$

This design is project-specific, but it is motivated by the idea that pathology-related information can be separated from background content in medical images (Xia, Chartsias and Tsaftaris, 2020).

GSTE then uses the mask to guide transformer tokens. If t_i is token i and w_i is its normalised mask-derived weight, the guided token is:

$$\tilde{t}_i = w_i t_i.$$

Hybrid A applies this guidance to a fixed 7×7 CNN-derived token grid. Hybrid B applies stronger guidance to raw-image patch tokens and the CNN descriptor. The ablation models remove PFD and GSTE while keeping the corresponding family structure.

3 Experimental Design

Four variants were trained and evaluated:

- **Hybrid A**: weaker PFD-GSTE guidance on CNN-derived tokens;
- **Hybrid B**: stronger PFD-GSTE guidance on raw-image patch tokens and CNN descriptor;
- **Ablation A**: A-family architecture without PFD-GSTE;
- **Ablation B**: B-family architecture without PFD-GSTE.

All models used the same fixed train, validation, and test splits. Training used AdamW, differential learning rates for pretrained CNN and transformer/fusion parameters, cosine annealing, label smoothing, gradient clipping, and validation macro-F1 early stopping. The best checkpoint for each model was selected using validation macro-F1 and then evaluated once on the held-out test set.

Evaluation used accuracy, macro-F1, weighted-F1, specificity, Cohen's kappa, Matthews Correlation Coefficient, confusion matrices, Grad-CAM++ for the CNN branch, Attention Rollout for the transformer branch, and Monte Carlo dropout uncertainty. Grad-CAM++ was used because it is designed for convolutional feature-map explanations (Chattopadhyay et al., 2018). Attention Rollout was used to inspect transformer-side token influence (Abnar and Zuidema, 2020). Monte Carlo dropout was used as a practical uncertainty estimate (Gal and Ghahramani, 2016).

4 Results

All models achieved strong held-out classification performance. Test accuracy ranged from 0.9852 to 0.9922. Ablation B achieved the best numerical performance, while Hybrid B produced the clearest tumour-centred Grad-CAM++ evidence in the reported qualitative cases.

Model	Accuracy	Loss	Macro-F1	Errors
Hybrid A	0.9875	0.0803	0.9875	16
Hybrid B	0.9852	0.0753	0.9849	19
Ablation A	0.9875	0.0847	0.9873	16
Ablation B	0.9922	0.0668	0.9920	10

Table 1: Held-out test performance across PFD-GSTE variants and matched ablations.

The numerical result favoured Ablation B, suggesting that the direct B-family RViT-style patch-token design was strongest for classification on this benchmark. However, explainability showed a different pattern. Hybrid B produced more tumour-centred Grad-CAM++ evidence in the selected cases, suggesting that stronger guidance affected visual evidence quality even when it did not improve final accuracy.

Uncertainty analysis showed that some wrong predictions remained highly confident. The clearest example was a pituitary image misclassified as meningioma by all four models. This indicates that the models often learned strong tumour evidence, but not always subtype-specific evidence. The uncertainty signal was therefore most informative when interpreted alongside the explanation maps.

5 Discussion

The main finding is not that PFD-GSTE universally improves accuracy. It does not. The strongest numerical model was Ablation B. Instead, the contribution of PFD-GSTE appears more clearly in evidence behaviour: stronger guidance helped the CNN branch produce more tumour-centred Grad-CAM++ maps in the reported examples.

This distinction is important. Shortcut learning is not only a question of prediction error; it is a question of evidence. A model may be accurate but visually untrustworthy, or slightly less accurate but more pathology-focused. In this study, Hybrid B represents the latter pattern more strongly than the other models.

The results also suggest that guidance strength must be controlled carefully. Stronger guidance can focus the model on suspicious regions, but it may also reduce surrounding context needed for subtype separation. This may explain why Hybrid B improved localisation evidence but did not outperform Ablation B numerically.

6 Limitations

This work has several limits. It uses one aggregated public benchmark and does not include external validation. It performs classification without segmentation masks, so localisation quality could not be measured through quantitative overlap with expert tumour annotations. The explainability analysis is qualitative and case-based. Monte Carlo dropout provides a practical uncertainty estimate, but calibration was not evaluated separately. The work should therefore be understood as a research prototype and not as a clinical system.

7 Future Work

A natural next step is to evaluate PFD-GSTE on external datasets and on multi-slice, multi-sequence, or three-dimensional MRI inputs. This would test whether the guidance remains useful when data distribution and anatomical context change. Future work should also compare explanation maps against segmentation masks where available, allowing quantitative localisation analysis.

A longer-term research aim is to develop PFD-GSTE as a reusable library for medical imaging classification tasks where segmentation labels are absent, expensive, or incomplete. Such a library could provide pathology-guided masking, token guidance, branch-specific explainability hooks, and uncertainty analysis as modular components for classification pipelines.

Another direction is to study multi-agent reinforcement learning for reliability-aware medical imaging workflows. Separate agents could specialise in classification, explanation assessment, uncertainty checking, and failure-case review. This would move the project beyond a single classifier toward a broader framework for evidence-aware medical AI evaluation.

8 Conclusion

This work proposed PFD-GSTE as an experimental guidance mechanism for hybrid CNN–Transformer brain tumour MRI classification without segmentation supervision. The study shows that all variants achieved strong held-out accuracy, but the most important result is more nuanced: the best numerical classifier was Ablation B, while the strongest tumour-centred CNN evidence appeared in Hybrid B. The evidence supports partial mitigation of shortcut-learning risk, mainly through improved visual focus rather than improved classification performance. The project therefore motivates future work on stronger validation, quantitative localisation, calibration, and reusable pathology-guided model components for medical imaging research.

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